

Math 1227 Student Worksheet

Vector, Matrix & Ordinary Differential Equations

RUET – Department of Mechatronics Engineering

Name: _____

Date: _____

Student ID: _____

Section: _____

Part A: Vector Analysis

A1. Vectors, Curves, and Differential Operators

Q1. Determine whether the vectors $a = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, $b = \mathbf{i} - 4\mathbf{k}$, $c = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ are linearly dependent or independent. If dependent, find a relation among them and show that their terminal points are collinear.

Marks: 4-5

Source: 2014-Q1(a), 2015-Q4(a), 2019-Q1(a)

Q2. Determine whether the vectors $r_1 = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $r_2 = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, $r_3 = -2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, $r_4 = 3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$ are linearly dependent or independent.

Marks: 3-4

Source: 2020-Q1(a), 2021-Q1(a)

Q3. Determine the volume of a tetrahedron whose adjacent edges are $A = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, $B = \mathbf{i} - 4\mathbf{k}$, $C = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.

Marks: 4

Source: 2020-Q1(b)

Q4. Find the unit tangent vector at $t = 2$ for the curve $x = t^2 + 1$, $y = 4t - 3$, $z = 2t^2 - 6t$.

Marks: 3-4

Source: 2019-Q1(b), 2023-Q1(b)

Q5. Find the curvature κ for the curve $x = t$, $y = t^2$, $z = \frac{2t^3}{3}$.

Marks: 3

Source: 2023-Q1(c)

Q6. What is the physical significance of $\nabla \cdot (\nabla \phi)$ (Laplacian)?

Marks: 4

Source: 2014-Q2(a)

Q7. Write down the geometrical interpretation of $\nabla \Phi$.

Marks: 5

Source: 2023-Q2(a)

Q8. Find the angles which the vector field $F = (y^2x - z)\mathbf{i} + (yz^2 + x)\mathbf{j} - 2x^2z^2\mathbf{k}$ makes with the coordinate axes. Also find $\nabla(\nabla \cdot F)$ at $(2, -1, 0)$.

Marks: 4

Source: 2014-Q1(b), 2014-Q1(c)

Q9. If $V = (x + 3y)\mathbf{i} + (y - 2z)\mathbf{j} + (x + az)\mathbf{k}$, determine the constant a so that the divergence of V is zero.

Marks: 4

Source: 2016-Q3(c)

Q10. Find constants a, b, c so that $V = (x + 2y + az)\mathbf{i} + (bx - 3y - z)\mathbf{j} + (4x + cy + 2z)\mathbf{k}$ is irrotational.

Marks: 3

Source: 2015-Q4(c)

A2. Directional Derivatives and Potentials

Q11. Prove that: (i) $\nabla \times (\nabla \phi) = 0$ and (ii) $\nabla \cdot (\nabla \times A) = 0$.

Marks: 4

Source: 2018-Q3(b)

Q12. Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$.

Marks: 4

Source: 2016-Q3(b), 2017-Q3(b)

Q13. Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ in the direction $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$.

Marks: 3

Source: 2021-Q1(b)

Q14. From the point $(1, 3, 2)$, in what direction is the directional derivative of $\phi = 2xz - y^2$ maximum? Find the magnitude of this maximum.

Marks: 3

Source: 2022-Q1(b)

Q15. Show that $A = (6xy + z^3)\mathbf{i} + (3x^2 - z)\mathbf{j} + (3xz^2 - y)\mathbf{k}$ is irrotational. Find ϕ such that $A = \nabla \phi$.

Marks: 4

Source: 2017-Q3(a)



Q16. Show that $F = (2xy + z^2)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$ is a conservative force field, and find the scalar potential.

Marks: 4-6

Source: 2020-Q2(a), 2022-Q1(c)



Q17. Show that $F = e^x \cos z \mathbf{i} + 2y \mathbf{j} - e^x \sin z \mathbf{k}$ is irrotational. Find scalar function ϕ such that $F = \nabla \phi$.

Marks: 4

Source: 2020-Q1(c)



Q18. Show that $F = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is conservative. Find scalar potential Φ such that $F = \nabla \Phi$.

Marks: 4

Source: 2021-Q1(c)



A3. Integral Theorems and Applications

Q19. If $F = 3xy \mathbf{i} - y^2 \mathbf{j}$, evaluate $\int_C F \cdot d\mathbf{r}$ where C is $y = 2x^2$ from $(0, 0)$ to $(1, 2)$.

Marks: 4

Source: 2014-Q2(b)



Q20. If $F = (2x + y)\mathbf{i} + (3y - x)\mathbf{j}$, evaluate $\int_C F \cdot d\mathbf{r}$ where C is the broken line from $(0, 0)$ to $(2, 0)$ to $(3, 2)$.

Marks: 5

Source: 2022-Q2(a)



Q21. Evaluate $\int_C A \cdot d\mathbf{r}$ along $x^2 + y^2 = 1$, $z = 1$, from $(0, 1, 1)$ to $(1, 0, 1)$ if $A = (yz + zx)\mathbf{i} + xz\mathbf{j} + (xy + 2z)\mathbf{k}$.

Marks: 4-5

Source: 2023-Q2(b), 2024-Q1(c)



Q22. Evaluate $\iint_S F \cdot \hat{n} dS$ for $F = x\mathbf{i} + 2y\mathbf{j} - z\mathbf{k}$ where S is the part of plane $2x + y = 6$ in the first octant cut by $z = 4$.

Marks: 4

Source: 2014-Q2(c)



Q23. If $F = 4xz\mathbf{i} - y^2\mathbf{j} + yz\mathbf{k}$, evaluate $\iint_S F \cdot \hat{n} dS$ over the surface of cube $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$.

Marks: 5-6

Source: 2016-Q4(b), 2018-Q4(a), 2020-Q2(b)



Q24. If $F = (2x^2 - 3z)\mathbf{i} - 2xy\mathbf{j} - 4x\mathbf{k}$, evaluate $\iiint_V (\nabla \times F) dv$, where V is bounded by $x = 0$, $y = 0$, $z = 0$, and $2x + 2y + z = 4$.

Marks: 5-6

Source: 2019-Q2(a), 2024-Q2(b)



Q25. Verify Green's theorem for $\oint_C \{(xy - x^2) dx + x^2y dy\}$ where C is the triangle bounded by $x = 1$, $y = 0$, and $y = x$.

Marks: 6

Source: 2016-Q4(a)



Q26. Verify Stokes' theorem for $A = 3y\mathbf{i} - xz\mathbf{j} + yz^2\mathbf{k}$, where S is $2z = x^2 + y^2$ bounded by $z = 2$, with C traversed clockwise.

Marks: 5

Source: 2023-Q3(b)

Q27. State and prove the Divergence theorem.

Marks: 5-6

Source: 2020-Q3(a), 2021-Q3(a), 2022-Q3(a), 2024-Q3(a)

Part B: Matrices

Q28. Apply the Divergence theorem to compute $\iint_S u \cdot \hat{n} \, ds$ where S is the cylinder $x^2 + y^2 = a^2$ between $z = 0$ and $z = b$, and $u = x\mathbf{i} - y\mathbf{j} + z\mathbf{k}$.

Marks: 6

Source: 2017-Q4(a)

B1. Definitions and Fundamental Properties

Q29. Define: square matrix, diagonal matrix, upper triangular matrix, symmetric matrix, and Hermitian matrix.

Marks: 3

Source: 2015-Q1(a)

Q30. Define symmetric, transpose, orthogonal, and idempotent matrices.

Marks: 4

Source: 2017-Q1(a)

Q31. Define matrix and symmetric matrix. If A and B are symmetric $n \times n$ matrices, prove that AB is symmetric if and only if A and B commute.

Marks: 4

Source: 2023-Q4(a)

Q32. Show that every square matrix can be uniquely expressed as the sum of a symmetric matrix and a skew-symmetric matrix.

Marks: 3

Source: 2022-Q4(b)

Q33. Find the symmetric and skew-symmetric parts of $A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$.

Marks: 3

Source: 2024-Q4(b)

Q34. What is the transpose of a matrix? Prove that $(AB)^T = B^T A^T$ for any compatible matrices A and B .

Marks: 4-5

Source: 2015-Q1(c)

Q35. What is an orthogonal matrix? Show that $\begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ is orthogonal.

Marks: 4

Source: 2016-Q1(a), 2018-Q1(a), 2023-Q4(c)

Q36. What is the adjoint of a matrix? Prove that the adjoint of a symmetric matrix is also symmetric.

Marks: 4

Source: 2024-Q4(a)

Q37. Prove that matrix multiplication is distributive over matrix addition.

Marks: 3

Source: 2015-Q1(b)



Q38. Prove that $(AB)^{-1} = B^{-1}A^{-1}$ for two non-singular matrices A and B .

Marks: 4

Source: 2017-Q1(b)



B2. Inverse and Rank of a Matrix

Q39. Prove that a square matrix A has an inverse if and only if A is non-singular.

Marks: 4

Source: 2015-Q2(b)

Q40. Find the inverse of $\begin{bmatrix} 1 & 2 \\ 5 & 11 \end{bmatrix}$ by elementary transformations.

Marks: 4-6

Source: 2014-Q5(b)

Q41. Find the inverse of $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$.

Marks: 4

Source: 2015-Q3(a)

Q42. What is an inverse of a matrix? Find the inverse of $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$.

Marks: 6

Source: 2019-Q3(a), 2020-Q4(c)

Q43. Find the adjoint and hence inverse of $\begin{bmatrix} 1 & 2 & 1 \\ 5 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$.

Marks: 4

Source: 2021-Q4(c)

Q44. Find the rank of $A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 1 & -1 & 4 & 0 \\ 2 & 2 & 8 & 0 \end{bmatrix}$ by reducing to normal form.

Marks: 5

Source: 2014-Q4(b), 2018-Q1(b)

Q45. Reduce matrix C to echelon form and find its rank, where $C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$.

Marks: 4

Source: 2016-Q2(a)

Q46. Define rank of a matrix. Reduce $A = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$ to normal form and find rank.

Marks: 6

Source: 2020-Q5(a)

Q47. Define rank of a matrix. Reduce $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ to normal form and find rank.

Marks: 5 *Source: 2022-Q4(c), 2024-Q6(a)*

B3. Eigenvalues and Characteristic Theory

Q48. Verify the Cayley-Hamilton theorem for $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.

Marks: 6

Source: 2019-Q4(b)

Part C: Ordinary Differential Equations

C1. Formation and Exact Differential Equations

Q49. Define differential equation. Give examples and mention some applications.

Marks: 2-4

Source: 2016-Q6(a), 2022-Q6(a), 2024-Q6(b)

Q50. Form a differential equation from the relation $Ax^2 + By^2 = 1$.

Marks: 4

Source: 2017-Q5(a)

Q51. Define differential equation with an example. Form the differential equation corresponding to $y = ae^{2x} + be^{-2x} + ce^x$, where a, b, c are parameters.

Marks: 4

Source: 2019-Q5(a)

Q52. Define exact differential equation. Show that a necessary condition for $Mdx + Ndy = 0$ to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Marks: 5-7

Source: 2014-Q6(a), 2014-Q6(b)

Q53. Define exact differential equation. Test whether $(x + y)^2 dx - (y^2 - 2xy - x^2) dy = 0$ is exact and solve it.

Marks: 4

Source: 2023-Q6(a)

Q54. What is exact differential equation? Find the general solution of $(12y + 4x^3 + 6x^2)dx + 3(x + xy^2)dy = 0$.

Marks: 3

Source: 2024-Q7(a)

Q55. Solve: $(2x - y + 1)dx + (2y - x - 1)dy = 0$.

Marks: 5-6

Source: 2014-Q6(c), 2019-Q5(b)

Q56. Define exact differential equation. Solve $y(1 + xy)dx = x(1 - xy)dy$.

Marks: 4

Source: 2016-Q5(b)

Q57. Solve: $(x^2 + 1) \sin y \, dx + x \cos y \, dy = 0$.

Marks: 3

Source: 2021-Q6(c)

Q58. Solve the initial value problem: $x \sin y \, dx + (x^2 + 1) \cos y \, dy = 0$, with $y(0) = \pi/2$.

Marks: 4

Source: 2022-Q6(b)

C2. Homogeneous, IF, and Linear First Order Equations

Q59. If $M(x, y)dx + N(x, y)dy = 0$ is homogeneous, show that substitution $y = vx$ transforms it into a separable equation in v and x .

Marks: 4

Source: 2018-Q5(b)

Q60. Solve: $\frac{dy}{dx} = \frac{2x + y - 2}{3x + y - 2}$.

Marks: 5

Source: 2017-Q5(b), 2019-Q5(b)

Q61. Define integrating factor. Find the IF of $\frac{dy}{dx} + y = 3x^2 - 3x$, and hence solve.

Marks: 5

Source: 2015-Q5(b)

Q62. What is integrating factor? Find the IF of $y dx + (3 + 3x - y) dy = 0$, and hence solve.

Marks: 4-6

Source: 2017-Q6(b), 2019-Q6(a)

Q63. Solve: $\frac{dy}{dx} + y = e^{-x}$.

Marks: 6

Source: 2014-Q7(a)

Q64. Find the general solution of $\frac{dy}{dx} + 2xy = 2e^{-x^2}$.

Marks: 4

Source: 2019-Q5(c)

Q65. Identify and solve: $(1 + x^2)\frac{dy}{dx} + 2xy = 4x^2$, with $y(0) = 1$.

Marks: 4

Source: 2023-Q7(c)

Q66. Solve $L\frac{di}{dt} + Ri = E \sin 2t$, where L, R, E are constants, and $i(0) = 0$.

Marks: 5-6

Source: 2018-Q6(a), 2024-Q8(b)

C3. Bernoulli and Higher Order Equations

Q67. Find the general solution of $\frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}$, with $y(1) = 2$.

Marks: 4

Source: 2015-Q5(c)

Q68. Solve $\frac{dy}{dx} + \frac{1}{3}y = \frac{1}{3}(1 - 2x)y^4$, with $y(0) = 1$.

Marks: 4-6

Source: 2016-Q6(c), 2018-Q6(b)

Q69. Solve: $\frac{d^2y}{dx^2} + y = \sin x$.

Marks: 6

Source: 2014-Q7(b)

Q70. Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10\sin x$.

Marks: 6

Source: 2015-Q7(b)

Q71. Solve $(x^2D^2 - 3xD + 4)y = x^2$, with $y(1) = 0$, $y'(e) = e^2$, where $D = \frac{d}{dx}$.

Marks: 5

Source: 2021-Q8(a)

Q72. Using variation of parameters, solve $\frac{d^2y}{dx^2} + (1 - \cot x)\frac{dy}{dx} - y \cot x = \sin^2 x$.

Marks: 6

Source: 2019-Q8(b)

